RPO Part 1 (Traveling to the KOZ)

Prompt

Using the Clohessy-Wiltshire (CWH) or Hill's equations to describe the relative motion of a chase vehicle (CV) approaching a target vehicle (TV) in orbit. MATLAB for computation and visualization. This is mean to simplify the problem instead of calculating the complexity of absolute orbits of both vehicles. CV is in a non-inertial coordinate frame fixed in the TV. The TV will Swift Observatory and the CV will represent SPEAR.

Reference Frame

- r (Radial-track): Parallel to the TV's radius vector from Earth.
- s (Along-track): Normal to the radial vector in the orbit plane.
- z (Cross-track): Perpendicular to the TV's orbital plane.

Assumptions

- TV's orbit is circular
  - Mean orbital rate is constant ($n \approx d\theta/dt$)
- CV and TV separation distance is much smaller than distance to Earth center

Discussion & Analysis

1. Development of a controller will be necessary to solve this problem; however, we are going to start with the uncontrolled EOMs. This means there will be CV coasting without thruster input $\rightarrow$ ($a_r = a_s = a_z = 0$) Which yields uncontrolled EOMs of the form:

$$\frac{d^2r}{dt^2} - 2n\frac{ds}{dt} - 3n^2r = 0$$
$$\frac{d^2s}{dt^2} + 2n\frac{dr}{dt} = 0$$
$$\frac{d^2z}{dt^2} + n^2z = 0$$

2. Initial values of the TV can be found Swift Observatory TLE history. SPEAR indents to rendezvous with Swift Observatory on August 22, 2026; so, values from that day will be used.
   - $R_{TV} = 6721.464$ km
3. Changing the along-track initial condition will simulate a V-bar approach; while changing the radial-track initial condition will simulate a R-bar approach.
   - $s_0 = -1000$ m: V-bar approach. For an uncontrolled simulation, the CV shares the same exact orbital period and velocity. This will result in the CV simply coasting and trailing the TV.
   - $r_0 = -1000$ m: R-bar approach. For an uncontrolled simulation, the CV has a faster orbital rate than the TV. This will result in the CV simply drifting past the TV.
4. What about placing the CV at an "above and ahead" location to the TV?
   - Higher orbit means the CV will be at a slower mean orbital rate than the TV. Because the CV is moving slower, it will naturally drift backward relative the the TV along-track axis. This allows the TV to naturally catch up to the CV from below.
   - The CV can enter a waiting period to allow the proper phasing to develop before executing the next maneuver.

Uncontrolled CWH Trajectory Simulation

Givens

- $\mu = 3.986004356 \times 10^{14}$ m^3/s^2
- $R_E = 6378.1366 \times 10^3$ m
- $R_{TV} = 6721.464 \times 10^3$ m

CV Initial Conditions

values subject to change

- $r_0 = 0$ or -1000 m
- $s_0 = 0$ or -1000 m
- $z_0 =$ m
- $\dot{r}_0 = 0$ or -1.0 m/s
- $\dot{s}_0 = 0$ or -1.0 m/s
- $\dot{z}_0 = 0$ m/s

Equations

$$n = \sqrt{\frac{\mu}{R_{TV}^3}} \text{ m/s}$$

$$T_{orbit} = \frac{2\pi}{n} \text{ sec}$$

Results

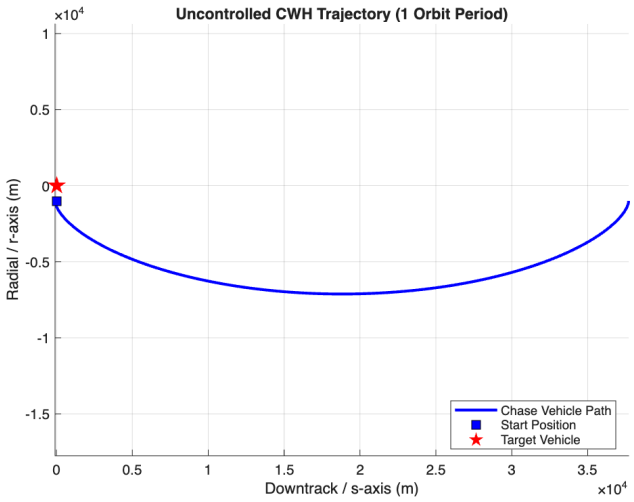


Figure 1: R-bar approach with initial conditions: $r_0 = -1000$ m and $\dot{r}_0 = 1.0$ m/s

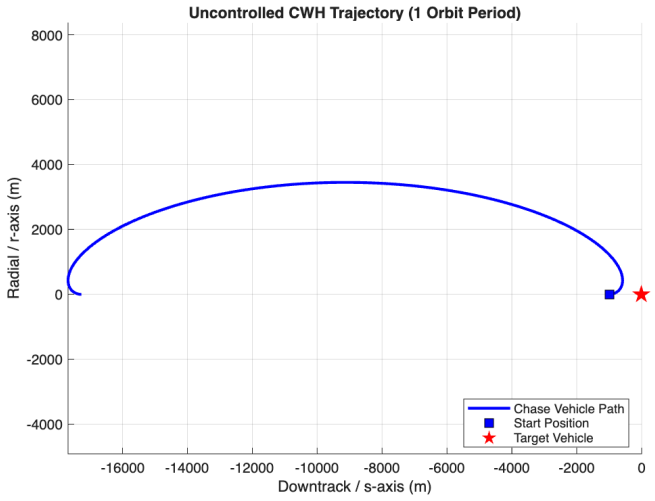


Figure 2: V-bar approach with initial conditions: $s_0 = -1000$ m and $\dot{s}_0 = 1.0$ m/s

Script Listing

Listing 1: CWH Trajectory without a Controller

```
% Elijah Bass
% AAE 450 - Orbit Boost
% Uncontrolled CWH Relative Motion Simulation

clear; clc; close all;

%% Given Orbital Parameters
mu = 3.986004356e14; % [m^3/s^2] Earth's gravitational parameter
R_TV = 6721.464e3; % [m] Target vehicle radius

%% Calculated Orbital Parameters
n = sqrt(mu / R_TV^3); % [rad/s] mean orbital rate
T_orbit = 2*pi/n; % [s] orbital period

%% Initial Conditions
r0 = -1000; % [m] radial track axis position
s0 = 0; % [m] along-track axis position
z0 = 0; % [m] cross-track axis position
rdot0 = 1.0; % [m/s] radial track axis velocity
sdot0 = 0; % [m/s] along-track axis position
zdot0 = 0; % [m/s] radial track axis position

%% Simulation Setup
X0 = [r0; s0; z0; rdot0; sdot0; zdot0]; % initial condition vector
t_span = [0, T_orbit]; % simulate for 1 orbital period
options = odeset('RelTol', 1e-8, 'AbsTol', 1e-8); % ode tolerances
[t, X_out] = ode45(@(t, X) cwh_unforced(t, X, n), t_span, X0, options);

%% Plotting
figure(1)
hold on; grid on;
plot(X_out(:,2), X_out(:,1), 'b-', 'LineWidth', 2, 'DisplayName', 'Chase Vehicle Path');
plot(s0, r0, 'ks', 'MarkerFaceColor', 'b', 'MarkerSize', 8, 'DisplayName', 'Start Position');
plot(0, 0, 'rp', 'MarkerFaceColor', 'r', 'MarkerSize', 12, 'DisplayName', 'Target Vehicle');
title('Uncontrolled CWH Trajectory (1 Orbit Period)');
xlabel('Downtrack / s-axis (m)');
ylabel('Radial / r-axis (m)');
legend('Location', 'best');
axis equal; hold off;
```

```
40
41 %% Unforced CWH Equations Function
42 function dX = cwh_unforced(~, X, n)
43
44     r = X(1); s = X(2); z = X(3);
45     rdot = X(4); sdot = X(5); zdot = X(6);
46
47     rddot = 2 * n * sdot + 3 * n^2 * r;
48     sddot = -2 * n * rdot;
49     zddot = -n^2 * z;
50
51     dX = [rdot; sdot; zdot; rddot; sddot; zddot];
52 end
```

Controller Development